

Viral Structure & Pathogenesis

Louise Thai, MD • 49-slide current deck • morphology → replication → pathogenesis → histopathology

8 official objectives

4 ScholarRx bricks

DNA vs RNA replication

viral histopathology

COVID-19 lecture model

SOURCE PRIORITY: the current 49-slide Thai deck controls lecture terminology, sequence, examples, and faculty emphasis. The matching MCG 2029 AI Notes handout (“Lecture 8: Viral Structure & Pathogenesis”) is used as secondary enrichment. When the handout adds board-style mnemonics, diagnostic details, or virus properties not shown on the current deck, they are labeled Handout-only enrichment.

Official Week 8 Objectives

1. Describe clinically relevant aspects of viral morphology.
2. Identify basic steps in the viral replication cycle targeted by antiviral drugs.
3. Compare DNA- and RNA-virus replication and apply it to pathology.
4. Recognize medically important DNA and RNA viruses, including COVID-19.
5. Describe modes of transmission of viral infections.
6. Discuss cellular and host immune responses to viral infections.
7. Relate laboratory diagnosis to virus structure and host-cell interactions.

8. Recognize clinical significance and histopathologic hallmarks of CMV, EBV, HSV, HPV, Rabies, Parvovirus B19, and RSV.

ASSIGNED SCHOLARRX

Virology Foundations & Frameworks

DNA Viruses Foundations & Frameworks

RNA Viruses Foundations & Frameworks

Cytomegalovirus & Epstein-Barr Virus

Slides 3-5, 14-15: What Makes a Virus a Virus?

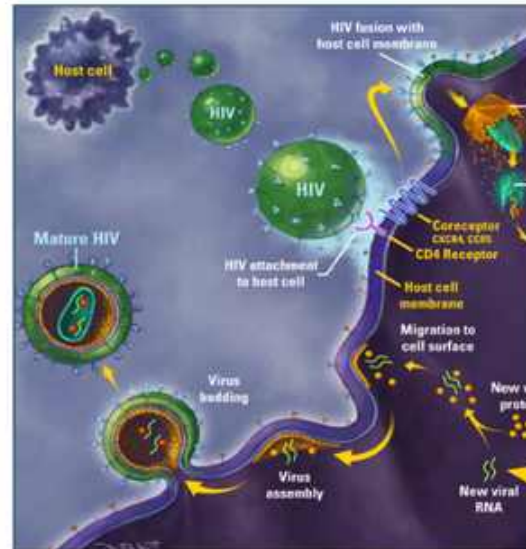
CORE DEFINITION

Virus = DNA or RNA genome + protein coat; obligate intracellular replication.

The extracellular particle is the **virion**, which transfers the viral genome to another susceptible cell.

Viral Envelope

- A portion of host **cell membrane**.
- Acquired from host cell during viral replication or release.
- Made up of phospholipids & protein/glycoprotein (**p/gp**) spikes.
- **P/GP** spikes play key roles in **binding** to host cells.



Slide 4. Envelope is acquired from host-cell membrane and contains phospholipids plus viral protein/ glycoprotein spikes used for host-cell binding.

HYI (High Yield Info)

Enveloped vs. Non-Enveloped Viruses

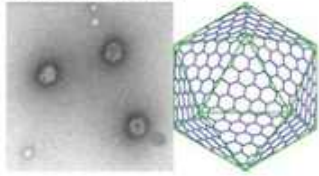
- Enveloped viruses are:
 - Fragile: Sensitive to acid, detergents, drying & heat
 - Sensitive to stomach acid and bile in the GI tract
 - Usually transmitted via body fluids
 - Not transmitted by fecal/oral route
 - Released from infected cells by budding or lysis
- Naked viruses are:
 - Stable: Resistant to acid, detergents, drying & heat
 - Resistant to stomach acid and bile in the GI tract
 - Readily transmitted by fecal/oral route
 - Released from infected cells by lysis

Slide 5. The deck's high-yield environmental distinction: envelopes are fragile; naked viruses are stable and tolerate acid, detergents, drying, and heat better.

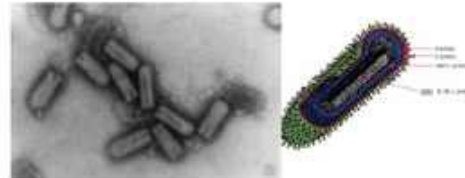
Feature	Enveloped	Naked / non-enveloped
Outer layer	Host-derived lipid membrane + viral spikes	Capsid is outermost layer
Environmental stability	Fragile: acid/detergent/drying/heat sensitive	Stable: relatively resistant
GI tract	Usually poor survival in acid/bile	Better survival in acid/bile
Transmission emphasis in deck	Usually body fluids; not fecal-oral	Readily fecal-oral
Release	Budding or lysis	Lysis

Virus Morphology

Icosahedral



Bulletshaped



Spherical



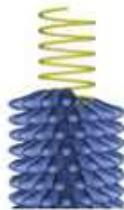
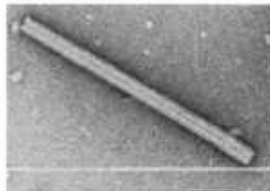
Ovoid



pleomorphic



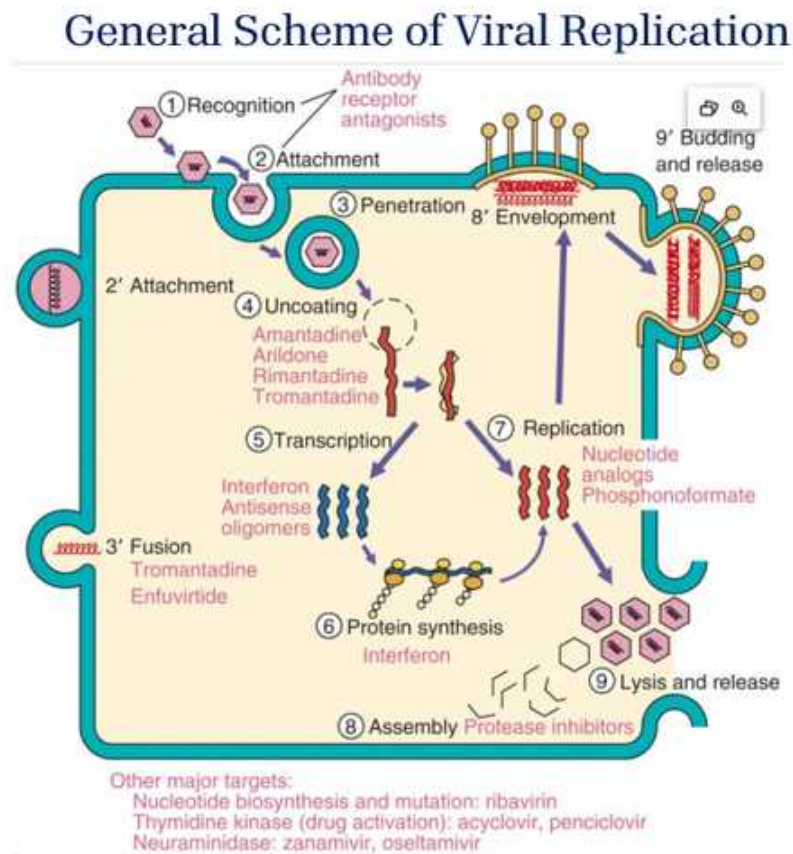
Filamentous



Slide 15. Morphologic forms shown in the current deck include icosahedral, spherical, ovoid, bullet-shaped, filamentous, and pleomorphic forms.

Handout-only enrichment: the AI handout explicitly names the **capsid** and **capsomeres** and reviews icosahedral vs helical capsid organization. That explanation helps interpret slide 15, but the current deck does not devote a text slide to capsid architecture.

Slides 6-9: Viral Replication Cycle = Antiviral Target Map



Slide 7. General replication scheme maps attachment, entry, uncoating, transcription/replication, protein synthesis, assembly, and release to druggable steps.

LIFE-CYCLE ORDER

Attachment → **entry** → **uncoating** → **genome replication / transcription** → **protein synthesis** → **assembly / maturation** → **release**.

Attachment: *HYI*

- Viruses recognize and bind to host cell surface molecules (**receptors**).
- Viral attachment proteins (**VAP**) determine host and tissue specificity.
- EBV binds to **CR2** on human B cells.
- HIV binds to CD4 & **chemokine receptors (CCR5, CXCR4)** on CD4+ T cells & macrophages
- Parvovirus B19 binds to globoside (blood group P antigen) on **proerythroblast**
- **SARS-CoV-2** binds to **ACE2**

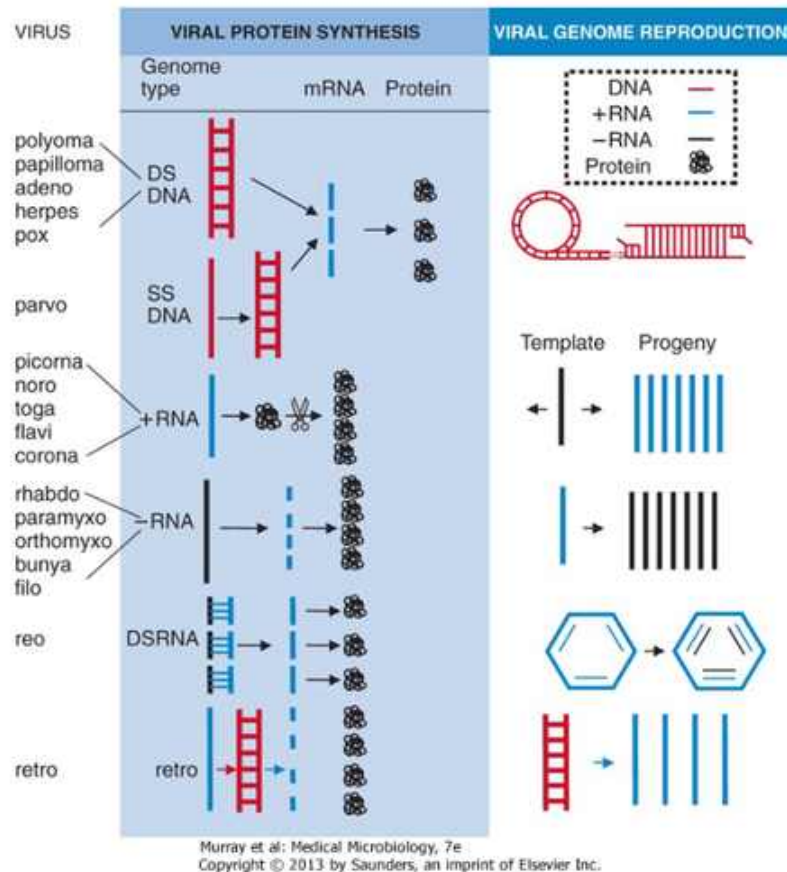
Slide 8. Attachment determines tropism. The current slide gives EBV → CR2 on B cells, HIV → CD4 plus CCR5/CXCR4, Parvovirus B19 → globoside/P antigen on proerythroblasts, and SARS-CoV-2 → ACE2.

Virus	Attachment receptor in current deck	Clinical logic
EBV	CR2	B-cell tropism
HIV	CD4 + CCR5/CXCR4	CD4 T cells / macrophages
Parvovirus B19	Globoside (P antigen)	Proerythroblast tropism
SARS-CoV-2	ACE2	Entry receptor emphasized again in COVID section

Handout-only enrichment: the 2029 handout adds concrete drug examples at each step (e.g., maraviroc for CCR5, enfuvirtide for fusion, amantadine for uncoating, polymerase/reverse-transcriptase inhibitors, protease inhibitors, oseltamivir for

influenza release). Use those as mechanism reinforcement, but the current Thai deck itself is more general at this point.

Slides 10-13: DNA vs RNA Replication



Slide 10. Genome-to-mRNA/protein overview across major viral genome classes. Use it conceptually rather than memorizing every line from the figure.

Replication of DNA Viruses: *HYI*

- Replication site is the **Nucleus** of host cell:
 - Exceptions: *poxvirus* and *hepatitis B virus* (HBV) replicate in the cytoplasm.
 - **HBV** uses an RNA intermediate to be copied back to DNA. Copying of RNA to DNA, as in retroviruses (e.g., HIV), is a function of RNA-dependent DNA polymerase = Reverse Transcriptase.
 - ***Nota Bene: Lamivudine inhibits Reverse Transcriptase, hence its Effectiveness against both HIV and HBV***

Slide 11. Current DNA-virus rule: replication is in the nucleus. The slide lists poxvirus and HBV as exceptions, and emphasizes HBV's RNA intermediate / reverse transcriptase.

Replication of RNA Viruses

- Replication site is the **Cytoplasm**:
 - Exceptions: Orthomyxoviruses (e.g., *influenza virus*) and retroviruses (e.g., *HIV*) replicate in the nucleus.
- RNA viral genomes: either **mRNA (+strand RNA)** or a template for mRNA (**-strand RNA**).
- RNA viruses use viral RNA-dependent RNA polymerase b/c cells have no means of replicating RNA.
- **Exception: Nota Bene**
 - Retroviruses (e.g., *HIV*) have RNA-dependent DNA polymerase (***Reverse Transcriptase***) that converts viral (+)ssRNA into dsDNA.

Slide 12. Current RNA-virus rule: replication is in the cytoplasm. Influenza/orthomyxovirus and retroviruses are nuclear exceptions. +RNA can function as mRNA; -RNA requires viral RNA-dependent RNA polymerase.

REPLICATION-LOCATION RULE

DNA → nucleus, except Pox (and the deck separately highlights HBV biology).
RNA → cytoplasm, except influenza and retroviruses.

Source-specific nuance: the current deck says HBV “replicates in the cytoplasm” while also emphasizing its RNA intermediate and reverse transcriptase. The companion handout phrases the standard board rule as “DNA viruses replicate in the nucleus except Pox,” then treats HBV as a special reverse-transcribing DNA virus. For this note, memorize the deck’s *HBV reverse-transcriptase exception biology* rather than forcing it into a one-line location mnemonic.

HYI:
Assembly of
Viruses
with
Segmented
Genomes

*Nota Bene: Influenza virus
& Rotavirus have segmental
genomes.*

Reassortment of 8
segments of influenza A
virus genome from different
animal species = ***Pandemic
Strain.***

Slide 13. Influenza and rotavirus have segmented genomes. The deck emphasizes influenza A reassortment of its 8 segments as the mechanism producing pandemic strains.

INFLUENZA

Antigenic drift = slow change → seasonal/yearly outbreaks. Antigenic shift = reassortment → potential pandemic.

Handout-only board enrichment: +ssRNA can be translated directly; -ssRNA must first be copied into +RNA and therefore needs viral RNA-dependent RNA polymerase available. The handout also supplies the “negative-sense viruses bring polymerase” mnemonic, which is not on the current deck.

Slides 16-18: Medically Important DNA & RNA Viruses

Medically Important Viruses

- **DNA (HHAPPPy) Viruses:**
 - *Herpesviridae*
 - *HSV, CMV, EBV*
 - *Hepadnaviridae*
 - Hepatitis B virus
 - *Adenoviridae*
 - *Adenovirus*
 - *Papillomaviridae*:
 - *HPV*
 - *Polyomaviridae*
 - JC virus, BK virus
 - *Parvoviridae*
 - *Parvovirus B19*
 - *Poxviridae*
 - Small pox virus
 - Vaccinia virus
 - *Molluscum contagiosum virus*
- **RNA Viruses:**
 - *Paramyxoviridae*
 - Measles, Mumps
 - Parainfluenza virus, *RSV*
 - *Orthomyxoviridae*
 - *Influenza virus*
 - *Rhabdoviridae*: Rabies virus
 - *Retroviridae*: *HIV*
 - *Reoviridae*: *Rotavirus*
 - *Picornaviridae*
 - Hep A virus, Poliovirus
 - *Togaviridae*: Rubella virus
 - *Flavoviridae*:
 - *Hepatitis C virus*
 - West Nile virus
 - *Bunyaviridae*: Hantavirus
 - *Caliciviridae*: Norwalk virus
 - *Coronaviridae*: *SARS-CoV-19*

Slide 16. The deck groups major DNA viruses using HHAPPPy and lists key RNA virus families/examples.

DNA family / virus	Deck examples / key point
Herpesviridae	HSV, CMV, EBV; latency emphasized on slide 17
Hepadnaviridae	HBV; hepatocellular carcinoma association
Adenoviridae	Respiratory disease, keratoconjunctivitis, gastroenteritis by serotype
Papillomaviridae	HPV 6/11 genital warts; HPV 16/18 cervical carcinoma
Polyomaviridae	JC, BK
Parvoviridae	B19; fetal anemia/hydrops via proerythroblast replication
Poxviridae	Smallpox, vaccinia, molluscum contagiosum

Unique Properties of RNA Viruses

- Orthomyxoviruses (e.g., *Influenza virus*):
 - Replicate in the nucleus
 - The genome consists of 8 segments
 - *Influenza A virus*:
 - Antigenic drift → slow changes → yearly outbreaks
 - Antigenic shift due to Reassortment → Pandemic
- Reoviruses (e.g., *Rotavirus*): (+/-) 11-segmented RNA:
 - Gastroenteritis in children; Prevention via vaccination
- Retroviruses (e.g., *HIV*): (+)RNA genome is converted to DNA that is integrated into host genome → oncogenic potential.

Slide 18. RNA-virus uniqueness slide: influenza 8 segments + drift/shift, rotavirus segmented RNA, and retroviral RNA → DNA integration.

Slide 16 typo note: the slide text appears to say “SARS-CoV-19” under Coronaviridae. The lecture objectives, cases, and subsequent slides clearly discuss **SARS-CoV-2 / COVID-19**; that is the terminology used in this companion.

Handout-only enrichment: the AI handout expands the family map with naked/enveloped status, + vs - sense examples, and additional viruses (e.g., rhinovirus, dengue, Ebola/Marburg). Those are useful board connections but are broader than the current deck’s own list.

Slides 20-21: Transmission & Progression of Disease

Transmission of Viral Diseases

- **Portal of Entry:** Broken skin (IV, bites); mucosa (eyes, mouth, RT, UG, GI).
- **Inhalation:** Influenza virus, *SARS-CoV-2* etc..
- **Replication:** In cells with receptors and means for synthesis.
- **Signs & Symptoms** accompany viral replication at the 1° site.
- **Viral Spread** via bloodstream and lymphatic system.
- Presence of virus in the blood is called *Viremia*.

Slide 20. Portal of entry includes broken skin and mucosal surfaces; inhalation examples include influenza and SARS-CoV-2. After local replication, viruses may disseminate by blood or lymph; virus in blood = viremia.

Progression of Viral Disease

- Acquisition ⇨ Initiation of infection at the 1° site
- Activation of innate defense mechanisms
- **Incubation Period:** Virus amplified ⇨ spread to 2° site
- **Prodromal Period:** Nonspecific S&S - fever & chills, body aches
- Replication ⇨ Specific **Clinical** manifestations:
 - Symptoms due to tissue damage and immune response
- Host immune response:
 - Limits disease progression
 - Contributes to **immunopathogenesis**
- Virus release from the tissue: **Contagion**
- Resolution or persistent infection/chronic disease:
Convalescence

Slide 21. Clinical progression: acquisition → innate defense → incubation/amplification → prodrome → specific disease → host response → shedding/contagion → resolution or persistence/chronic disease.

PROGRESSION FRAMEWORK

Acquisition → primary replication → dissemination/incubation → prodrome → tissue-specific disease → immune response → shedding → resolution or persistence.

Slides 22-24, 31: How Viruses Cause Injury

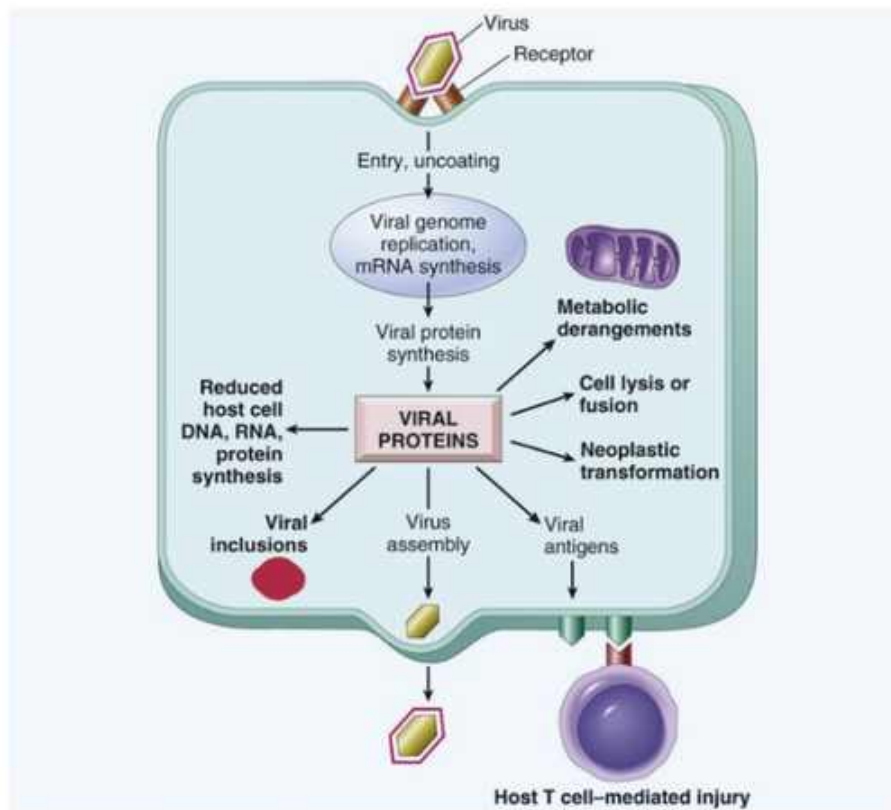


FIGURE 8-3 Mechanisms by which viruses cause injury to cells.

Slide 22. Viruses can reduce host DNA/RNA/protein synthesis, cause metabolic derangement, lysis/fusion, inclusions, neoplastic transformation, and host T-cell-mediated injury.

Histopathology

- **Syncytium:** Cell surface expression of viral GPs promotes fusion of cells ⇨ multinucleated giant cells (syncytium)
 - HSV, VZV, HIV, **RSV**, **SARS-CoV-2**
 - Syncytia help cell-to-cell spread and disguise the virus
- **Chromatin Margination:**
 - **Parvovirus B19**, HSV, Adenovirus
- **Inclusion bodies** in the **nucleus or cytoplasm:**
 - Result from changes in cell membrane or chromosome
 - Represent viral replication site or **capsids**
 - Staining pattern and location are diagnostic
- **Koilocytosis** (cytoplasmic vacuolization)
 - **HPV**

Slide 23. Current deck histopathology framework: syncytia, chromatin margination, inclusion bodies, and koilocytosis.

Pattern	Viruses named in current deck
Syncytium	HSV, VZV, HIV, RSV, SARS-CoV-2
Chromatin margination	Parvovirus B19, HSV, adenovirus
Inclusion bodies	Location/staining can be diagnostic
Koilocytosis	HPV

Immunopathology

- Inflammatory response initiated by cell-mediated immunity causes tissue damage.
- **Enveloped** viruses are notoriously associated with immunopathologic conditions (HBV, HCV, paramyxo- & orthomyxoviruses).
- Influenza virus: **Flu-like** symptoms due to interferon & cytokine production.
- Measles & mumps viruses: Delayed/T-cell-mediated (**type IV**) hypersensitivity.
- HBV & HCV: Immune complex, **type III** hypersensitivity.
- HIV: Immunosuppression

Slide 31. The lecture explicitly links viral disease to immunopathology: interferon/cytokines in influenza, delayed T-cell responses in measles/mumps, immune-complex mechanisms with hepatitis viruses, and immunosuppression in HIV.

Handout-only enrichment: the 2029 notes organize infection patterns as **acute, chronic, and latent**, and summarize interferons, NK cells, neutralizing antibodies, and CD8 T cells. That framework helps LO 6, but the current deck gives less dedicated text to these categories.

Slides 24-30, 32-34: Histopathology You Should Recognize

HYI: Diagnostic Cytopathology

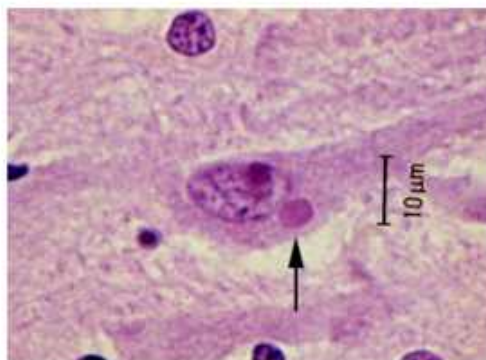
Intracellular Inclusions at the site of replication that represent **nucleic acids and proteins**:

- *Negri bodies* – **eosinophilic** intracytoplasmic inclusions in infected pyramidal cells, Ammon's horn: *Rabies virus*
- *Owl's eye* (**basophilic** nuclear inclusions): **CMV**
- *Cowdry bodies* – **eosinophilic** nuclear inclusions, with margination of chromatin: **HSV, VZV**
- *Margination of chromatin*: **Parvovirus B19**
- *Koilocytosis* (Cytoplasmic Vacuolization): **HPV**

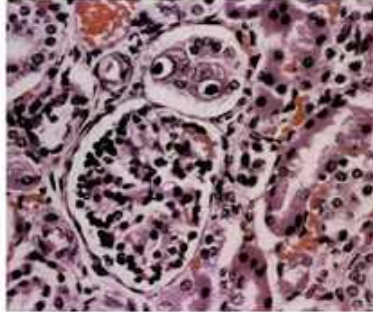
Slide 24. Faculty high-yield cytopathology list: Negri bodies (Rabies), owl's-eye nuclear inclusions (CMV), Cowdry nuclear inclusions with chromatin margination (HSV/VZV), Parvovirus B19 chromatin margination, and HPV koilocytosis.

Virus	Current-deck hallmark	Clinical association emphasized
Rabies	Negri bodies: eosinophilic intracytoplasmic inclusions in neurons	Pathognomonic morphology
CMV	"Owl's-eye" basophilic intranuclear inclusion	Disseminated infection; retinitis in AIDS
HSV / VZV	Cowdry eosinophilic intranuclear inclusions + chromatin margination	HSV encephalitis image shown
Parvovirus B19	Chromatin margination	Fifth disease; hydrops fetalis
HPV	Koilocytosis	Cervical dysplasia/carcinoma context
RSV	Syncytium formation	Pediatric pneumonia vignette

Rabies: Negri body in neuron (Pathognomonic)



CMV: Owl's-Eyes Intranuclear Inclusion Bodies

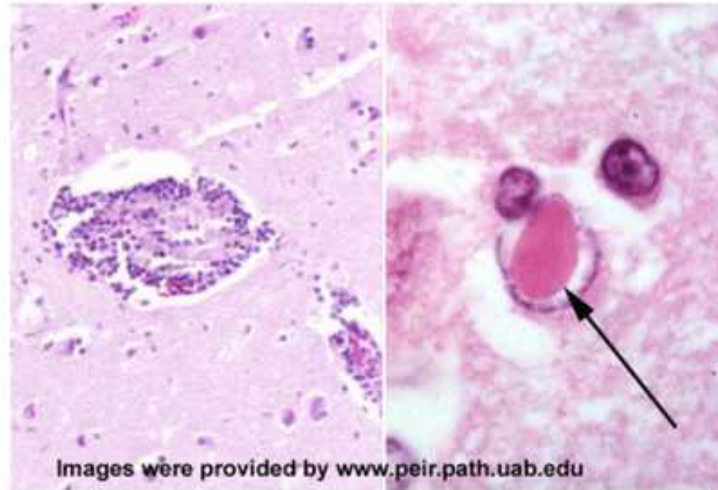


Cross-section of autopsied kidney from a 3 m/o boy who died of disseminated CMV infection contracted in utero. Renal tubule contains basophilic intranuclear viral inclusions.

Nota bene: CMV causes retinitis in patients with AIDS

Slide 26. CMV kidney example with basophilic intranuclear 'owl's-eye' inclusion.

HSV: Cowdry Inclusion Bodies



HSV encephalitis: *Acidophilic (eosinophilic)* inclusions surrounded by clear halos within nuclei; margination of the chromatin on nuclear membrane

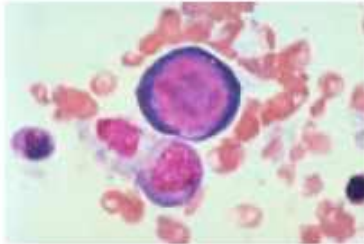
Slide 27. HSV Cowdry inclusions and chromatin margination.

Parvovirus B19

Erythema infectiosum
(Fifth Disease)



*Margination of
Chromatin*

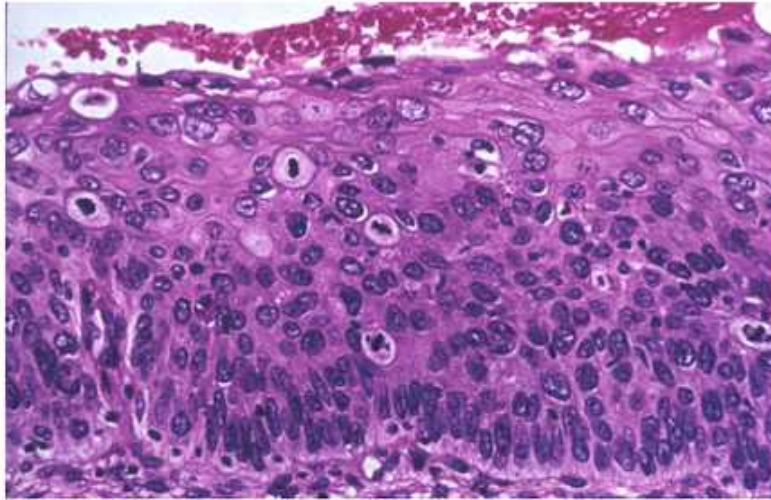


Hydrops fetalis



Slide 28. Parvovirus B19: fifth disease, chromatin margination, and hydrops fetalis are tied together.

HPV: Abnormal Cervical Biopsy w. *Koilocytosis*



Slide 29. HPV koilocytosis in an abnormal cervical biopsy.

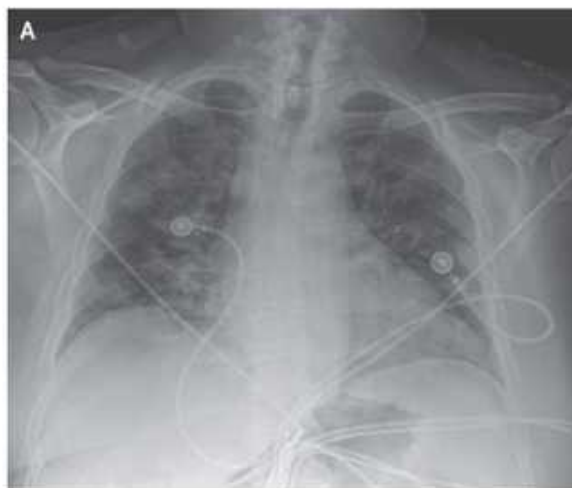
Ozzy checkpoint - click him once you can match Negri / owl's-eye / Cowdry / koilocyte / syncytium without looking.

Objective gap: EBV. The official objective asks for EBV histopathologic recognition, but the current deck does **not** show a dedicated EBV histology image. The matching 2029 handout adds **atypical lymphocytes (Downey cells) in peripheral blood** as its EBV recognition clue. Keep that labeled as handout enrichment rather than pretending it was shown in the current slide deck.

Handout-only diagnostic enrichment: the AI notes add PCR for viral nucleic acid, antigen testing, and serology (IgM vs IgG) as a general lab-diagnosis framework. The current deck's LO includes lab diagnosis, but it does not provide a dedicated general diagnostic-methods slide sequence.

CXR on Admission

- Bilateral, multifocal, patchy “ground-glass” opacities
- Presumptive Dx: Pneumonia



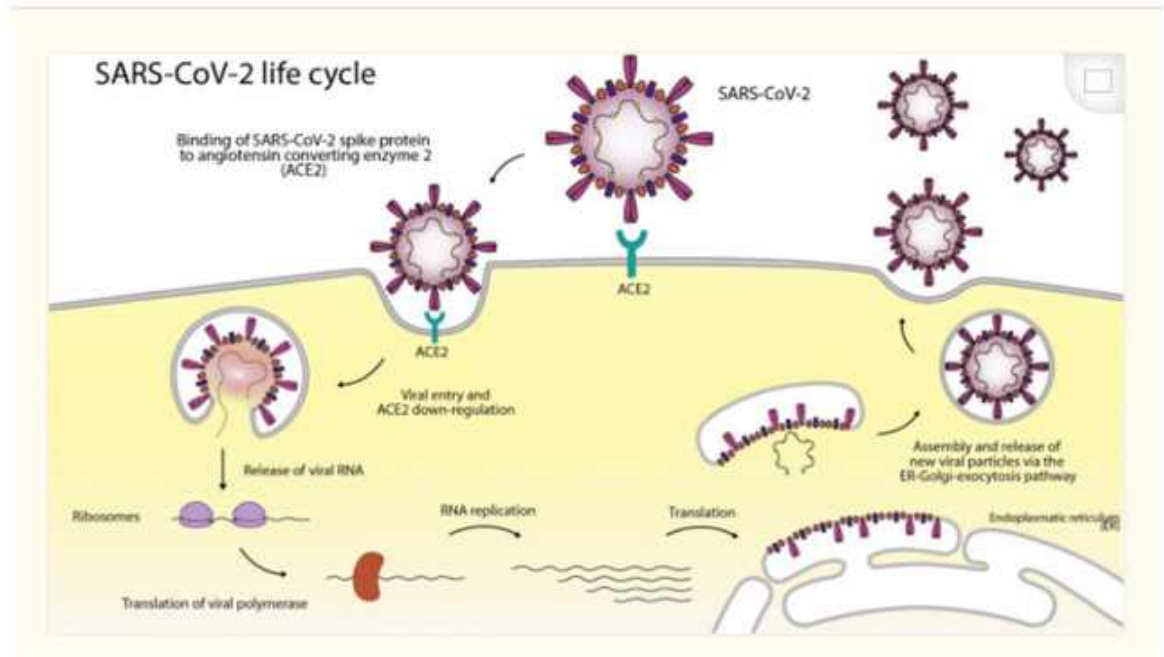
Slide 36. Admission CXR in the lecture case shows bilateral multifocal patchy ground-glass opacities.

SARS-CoV-2: Pathogenesis

- SARS-CoV-2 binds to **ACE2**, a peptidase that cleaves angiotensin II (**Ang II**) to Ang 1-7
- Binding of S1 to the ACE2 receptor $\Rightarrow$ cleavage of ACE2 $\Rightarrow$ **Loss of ACE2 function** $\Rightarrow$ **$\uparrow$ Ang II**
- Upregulation of angiotensin II (**Ang II**) leads to vasoconstriction, coagulopathy, alveolar damage and respiratory failure.
- Downstream effects of **Ang II**: ARDS, myocardial injury, renal failure $\Rightarrow$ High mortality among COVID-19 patients with CV & metabolic diseases

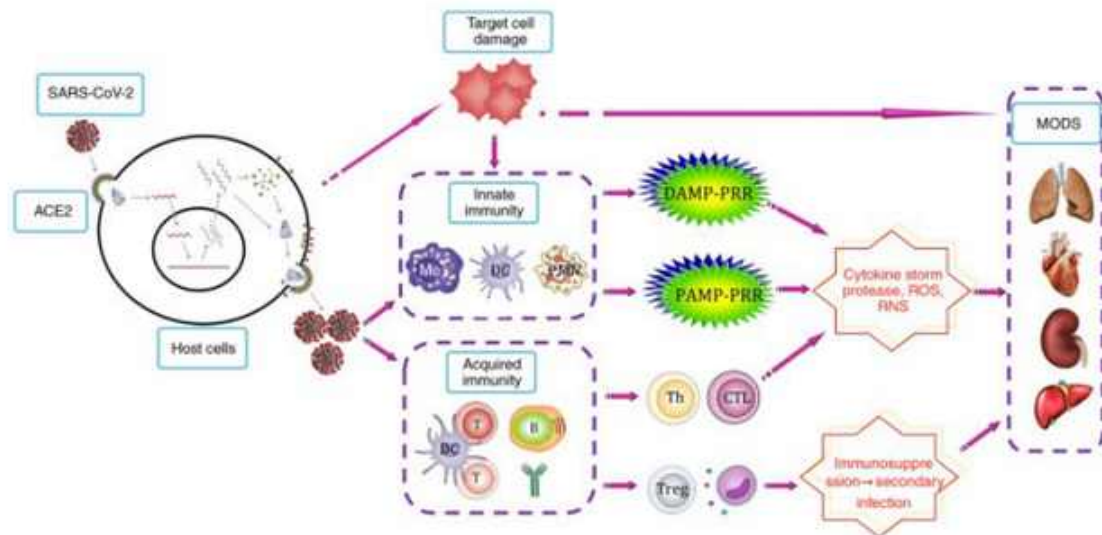
Slide 40. Lecture pathogenesis model: SARS-CoV-2 binds ACE2; loss of ACE2 function is presented as increasing Ang II, contributing to vasoconstriction, coagulopathy, alveolar damage, and respiratory failure.

Use this as the lecture model: SARS-CoV-2 $\rightarrow$ ACE2 binding / loss of ACE2 activity $\rightarrow$ $\uparrow$ Ang II $\rightarrow$ vascular and inflammatory injury $\rightarrow$ ARDS / myocardial injury / renal failure. This companion is preserving the current lecture's causal framing rather than substituting an outside mechanism.



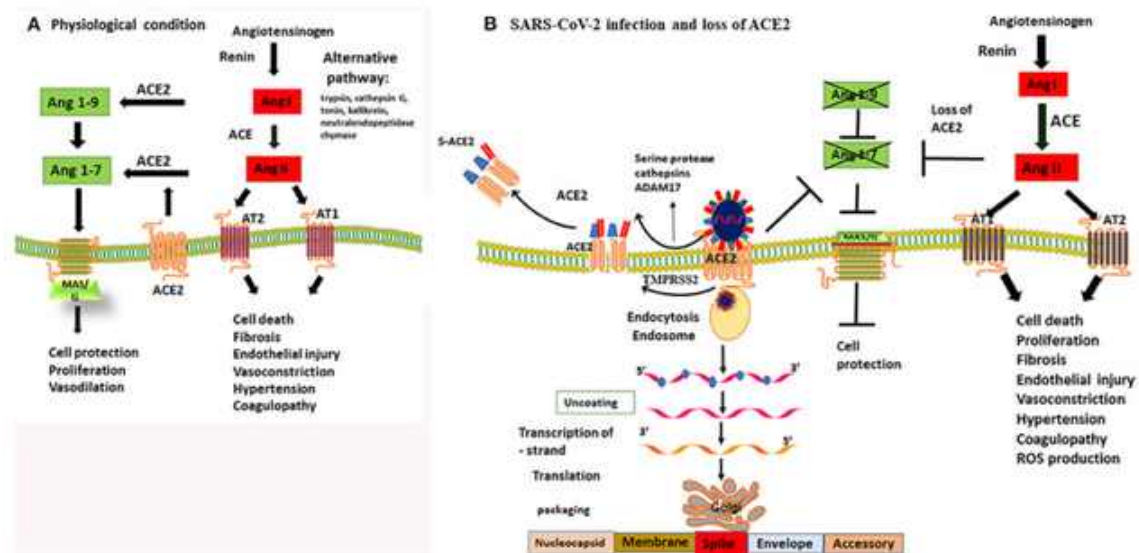
Slide 41. SARS-CoV-2 life-cycle figure: ACE2 binding/entry, RNA release, translation, RNA replication, ER-Golgi assembly, and particle release.

Pathogenesis of COVID-19



Slide 42. COVID-19 pathogenesis diagram connects direct target-cell damage, innate/adaptive responses, cytokine storm/ROS/RNS, immunosuppression/secondary infection, and multiorgan dysfunction.

Pathogenesis of *SARS-CoV-2* Infection



Slide 44. Expanded ACE2/RAAS schematic used by the lecture to explain loss of protective ACE2 signaling during infection.

Treatment / prevention named on slide 45

Agent	Mechanism stated in deck
Remdesivir	Inhibits SARS-CoV-2 RNA-dependent RNA polymerase
Nirmatrelvir / ritonavir	Nirmatrelvir = protease inhibitor; ritonavir = CYP3A4 inhibitor
Molnupiravir	Nucleoside-analog prodrug; deck describes competitive inhibition of RdRp
Vaccines	mRNA vaccines (Moderna, Pfizer-BioNTech); protein-subunit vaccine (Novavax)

Summary of HYI Points: Viral Pathogenesis

Virus	Diseases	Pathogenesis/Histopathology	
Parvovirus B19	- Fifth disease in peds <i>Hydrops fetalis</i> in neonates	Margination of chromatin	
RSV	Pneumonia in peds	Syncytium formation	
CMV	- Disseminated infection in immunocompromised - CMV retinitis in AIDS	Nuclear inclusions - owl's eyes	
SARS-CoV-2	- Pneumonia - ARDS - Systemic: multiple organ systems involved	- Binding to ACE-2 Receptors → Loss of ACE2 → ↑ Ang II - Cytokine storm - MODS	

Slide 48. Faculty final high-yield table links Parvovirus B19, RSV, CMV, and SARS-CoV-2 to their hallmark diseases/pathology.

High-Yield Traps & Vignette Recognition

If you see...	Think...	Reason
Fragile virus; body-fluid transmission; budding	Enveloped virus	Lipid envelope is environmentally sensitive
Fecal-oral survival / acid resistance	Naked virus	Stable capsid without lipid envelope
Segmented influenza A coinfection + pandemic	Antigenic shift	Genome reassortment
Slapped-cheek rash / hydrops / proerythroblast tropism	Parvovirus B19	P antigen receptor + erythroid precursor replication
Transplant/AIDS + retinitis + owl's-eye inclusion	CMV	Basophilic intranuclear inclusion
Neuron + eosinophilic cytoplasmic inclusion	Rabies	Negri body
Cervical lesion + perinuclear halo	HPV	Koilocytosis
Pediatric pneumonia + giant multinucleated cells	RSV	Syncytium formation
Enveloped +RNA virus binding ACE2	SARS-CoV-2	Lecture case definition

LAST 60 SECONDS

Envelope predicts stability/transmission. Genome predicts replication strategy. Receptor predicts tropism. Viral proteins + host immunity predict injury. Histology gives the buzzword diagnosis.

Source hierarchy: Thai 49-slide current deck is primary. Official Week 8 LOs define objectives/ScholarRx. The MCG 2029 AI Notes "Lecture 8: Viral Structure & Pathogenesis" handout is secondary enrichment and is explicitly labeled when it adds material not shown on the current deck.